Continuing Induction

Example

Given a collection of stamps

(Say unlimited supply of $3¢$ and $17¢$ stamps)

What totals can we attach to an envelope?

For instance $10¢$ is not a possible total.

Let $Q(n)$ be "It is possible to put exactly n cents on an envelope"

For these denominations,
- $Q(0)$ is true
- $Q(2)$ is false
- $Q(10)$ is false

Can we figure out out the values for which $Q(n)$ is true?

Observation

If $Q(k)$ is true, then $Q(k+3)$ and $Q(k+17)$ must be true.

If somebody tells you a recipe for paying k cents, you can pay $k+3$ cents by attaching that recipe, and then adding an extra $3¢$ stamp.

The reverse $Q(k+3) \to Q(k)$ is false, we can't take 3 cents off unless there was a $3¢$ stamp present

Example

$Q(17)$ is true,

but

$Q(14)$ is false

Approach 1

Find consectuive values, using trial and error, that are possible.

Note

$Q(34)$ is true $(2 * 17¢)$

$Q(33)$ is true $(11 * 3¢)$

$Q(32)$ is true $(1 * 17¢ + 5 * 3¢)$

We can extend this and conclude

$Q(n)$ is true whenever $n \geq 32$

to pay $35¢$, use the $32¢$ recipe plus one $3¢$ stamp

Extend this to $n+1 \mid n \geq 32$ ...

To keep the recipe rolling, we need to show that being able to pay consecutive values, lets us pay higher consecutive values.

Let $R(n)$ be "$Q(n) \wedge Q(n+1) \wedge Q(n+2)$"

1. $R(n) \to Q(n+1) \land Q(n+2)$ (Simplification)
2. $R(n) \to Q(n)$
3. $Q(n) \to Q(n+3)$

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4. $\therefore R(n) \to Q(n+3)$ (Hypothetical Syllogism)
5. $R(n) \to (Q(n+1) \land Q(n+2) \land Q(n+3))$
6. $\equiv R(n) \to R(n+1)$

Since n was generic

$\therefore \forall k \geq 1 R(n) \to R(n+1)$ (Universal Generalization)

We saw $R(32)$ (Direct proof (Team approach))

$\therefore \forall n \geq 32 R(n)$

and thus

$\forall n \geq 32 R(n)$

This is the best we can do.

$P(31)$ is false, so $R(31)$ is false,

and thus

$R(1), R(2), ...R(30)$ are all false

We still need to evaluate:

$Q(1), Q(2), ..., Q(31)$ By brute force.

Alternate Approach

Is there a situation when we <u>can</u> prove $Q(k) \to Q(k+1)$

Two Observations

- $6 \times 3¢ - 17¢ = 1¢$

Also

- $2 \times 17¢ - 11 * 3¢ = 1¢$

So if $k¢ = m * 3¢ + n * 17¢$

Then we can make $(k+1)¢$ as long as

$m \geq 11$ or $n \geq 1$

Obstacles

Are only possible if:

$m \leq 10$ and $n \leq 0$

In which case

$k¢ = n * 3¢ + n \times 17¢ \leq 10 * 3¢ + 0 \times 17¢ = 30¢$

Thus

If $k \geq 31$, the obstacles can't both happen. and so

$Q(k) \to Q(k+1)$

This direct proof shows:

$\forall k \geq 31 Q(k) \to Q(k+1)$

In fact $Q(31)$ is false, but $Q(32)$ is true

$\therefore \forall n \geq 32 Q(n)$

Case Analysis

To justify case analysis, with n cases, we rely on the (note yet proved in this course) claim that:

$(p_1 \to q) \land (p_2 \to q) \land ... \land (p_n \to q) \equiv (p_1 \lor p_2 \lor p_3 \lor ... \lor p_n) \to q$

an inifnite number of different claims, one claim or for each n

Let $P(n) = (p_1 \lor p_2 \lor ... \lor p_n) \to q$

We can use (modified) induction to prove,

$\forall n P(n)$

Base case $P(2)$ can be proved using a truth table.

Our goal is an expression involving,

$(p_1 \to q) \land (p_2 \to q) \land ... \land (p_n \to q) \land (p_{n+1} \to q)$

Let's start with that and track assumptions.

$\equiv ((p_1 \lor p_2 \lor ... \lor p_n) \to q) \land (p_{n+1} \to q)$ provided $P(n)$

$\equiv ((p_1 \lor p_2 \lor ...p_n) \lor p_{n+1} \to q)$

We conclude:

If $P(2)$ and $P(n)$ then

LHS of $P(n+1) \equiv$ RHS of $P(n+1)$

thus

$P(n) \land P(2) \to P(n+1)$ (Not quite right for induction)

Fix

Let $Q(n)$ be $P(n) \land P(2)$

thus

$Q(n) \to Q(n+1)$

and $Q(2)$